

Manuel Carlos Ramos Alves

SOFTWARE ENGINEER

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PROFILE

Software engineer working with Python and applied AI, with an MSc from FEUP, University of Porto. At CERN I built a Python + Flask optimization microservice that was merged into the platform the team uses, and at zerozero I build a GenAI content pipeline with LLM APIs, embeddings and pgvector. I also built and run worldcup2026ml.pt, a live machine-learning platform forecasting the FIFA World Cup with a Python model.

CORE SKILLS

Languages: Python · TypeScript · JavaScript · Java · C++ · C# · SQL · PHP · C

AI & ML: LLM APIs · RAG & vector search (pgvector) · NLP & embeddings · scikit-learn · pandas · NumPy · Stable-Baselines3 · SPADE (multi-agent) · Google OR-Tools (CP-SAT) · Apache Solr · Power BI

Back-end: Flask · Django · Node.js · Express.js · Spring Boot (REST APIs, Hibernate/JPA) · Laravel

Front-end: React · Next.js · Angular · Material UI · Bootstrap · Three.js · HTML & CSS

Databases: PostgreSQL (Supabase, pgvector) · Oracle · SQLite · CouchDB

DevOps & Tools: Git · Docker · GitHub Actions (CI/CD) · GitLab · Linux & Bash · Figma · Unity

PROFESSIONAL EXPERIENCE

zerozero (ZOS, Lda.) | Full-Stack Developer

Jul 2026 – Present

Vila Nova de Gaia, Portugal

- Built a GenAI content pipeline for the company's sports newsrooms in Node + TypeScript + Express over PostgreSQL/pgvector: article ingestion, topic clustering with embeddings, entity extraction and source-grounded draft generation behind a provider-agnostic LLM layer.
- Delivered a match-photography platform in React + TypeScript over an Express + SQLite back-end, with resumable uploads over the tus protocol and scheduled two-way synchronization with the internal and Getty APIs.

CERN | MSc Thesis Student (Cooperation Associate)

Sep 2025 – Jul 2026

Geneva, Switzerland (remote phase from Porto; on-site March – May 2026)

- Developed the MSc thesis "Intelligent Planning Optimization for Programmed Stops at CERN": state-of-the-art review on constraint programming and resource-constrained project scheduling (RCPSP), requirements gathering with CERN stakeholders, and end-to-end system architecture.
- Extended CERN's internal Scheduling Tools platform with a React + TypeScript + Material-UI front-end (interactive Gantt chart, template engine and schedule editor with FS/SS/FF/SF dependencies and MS Project-style constraints) over a Spring Boot + Oracle back-end.
- Built a Python + Flask microservice using Google OR-Tools (CP-SAT) for constraint-based schedule optimization (working-time calendars, CERN holidays, team/location concurrency limits), with MSPDI XML export for Microsoft Project interoperability; both the front-end and the optimization service were merged into the platform in use by the EN-ACE team.

Pontual - IT Business Solutions | Software Developer Intern

Feb 2024 – Jun 2024

Santa Maria da Feira, Portugal

- Developed a procurement web portal in Angular + TypeScript + Bootstrap, integrated with the Cegid PRIMAVERA ERP for data synchronization.
- Implemented multi-level approval workflows, team management and hierarchical role-based permissions for orders and expenses, plus a dashboard landing page with interactive charts.
- Designed UI prototypes in Figma and produced flow and architecture diagrams documenting the end-to-end procurement process.

Other Experience | Part-time & seasonal alongside studies

2019 – 2026

- Sports Coordinator at the AMB Volleyball Cup (seasonal, 2019 – 2026), running on-court operations for a 6,000+ athlete tournament: team check-ins, volunteer and referee deployment, live scheduling and score reporting.
- Café server (Futpark, 2022 – 2026) and pizza-delivery driver (2021 – 2022), part-time throughout my studies; volunteer at Web Summit 2024, Lisbon.

EDUCATION

MSc in Informatics and Computing Engineering | Faculty of Engineering, University of Porto Sep 2024 – Jul 2026
Final grade: 17/20 · Dissertation: 19/20 — “Intelligent Planning Optimization for Programmed Stops at CERN”, developed at CERN, Geneva.

BSc in Informatics and Computing Engineering | Faculty of Engineering, University of Porto Sep 2021 – Jun 2024
Final grade: 15/20 · Capstone: procurement web portal with ERP integration, developed at Pontual.
Initially enrolled in the integrated MSc in Electrical and Computer Engineering at the same faculty (2019 – 2021) before switching to Informatics.

PROJECTS

World Cup 2026 ML Predictor | Personal project, live at worldcup2026ml.pt Jun – Jul 2026

- Built and deployed a live ML web app forecasting the FIFA World Cup 2026: Poisson + Elo model over 150 years of results, with million-scale Monte Carlo simulation for title odds, brackets and Golden Boot projections (Python, pandas, NumPy); its final predictions proved highly accurate across the tournament.
- Added “Beat the Machine”, a live prediction game scored against the model, built with Google OAuth + Supabase (PostgreSQL/RLS), a vanilla-JS dashboard and a GitHub Actions pipeline that ingests live data and redeploys every 2 hours.

Multi-Agent Waste Collection System | Agents and Multi-Agent Systems Feb – Jun 2025

- Simulation of city waste collection with SPADE agents over XMPP (Openfire): bin, truck and world agents that coordinate only by exchanging messages, shown on a real map through a Flask web interface, across three scenarios.

Deep Reinforcement Learning Agents | Agents and Multi-Agent Systems Feb – Jun 2025

- Trained and compared A2C, DQN and PPO agents on LunarLander with Stable-Baselines3, testing changes to learning rate, batch size, exploration and PPO clipping, with training logs, episode videos and a comparison notebook.

Sentiment-Based Recommender & Community Forecasting | Machine Learning Complements Feb – Jun 2025

- Sentiment-enhanced recommender on Yelp data (collaborative filtering with KNN/SVD, VADER sentiment, TF-IDF/Word2Vec embeddings), plus community-evolution forecasting using graph-based community detection and Holt-Winters time-series smoothing.

Movie Information Retrieval System | Information Processing and Retrieval Sep 2024 – Jan 2025

- Built a search engine over a custom IMDb dataset scraped and cleaned in Python; indexed in Apache Solr with basic and boosted schemas and evaluated with TREC metrics (precision/recall).
- Extended it with semantic search via sentence-transformer embeddings behind an Express.js web interface.

NoSQL Game Store Platform | Non-Relational Databases Feb – Jun 2025

- Steam-like game store built as a full-stack app with a React/TypeScript front-end, a Flask back-end and CouchDB for document storage, featuring authentication, a Recharts analytics dashboard, Docker Compose deployment, and custom map/reduce views and aggregations across users, games, orders and carts.

LANGUAGES

Portuguese	Native or bilingual proficiency
English	Full professional proficiency (C2)
Spanish	Elementary proficiency (A2)
French	Elementary proficiency (A2)

INTERESTS

Volleyball: 2× Portuguese national champion (U16 2015/16, U21 2020/21), with 10+ years of federated competition.